SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

QUESTIONS: 5 **Total Marks:50**

84 %

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I K·Sudhesika (192511070) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding(2)	Identifies most errors with minor gaps(1.5)	Identifies some errors but misses key issues(1)	Unable to identify errors correctly(0)
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues(2.5)	Uses appropriate debugging methods with minor errors(2)	Limited debugging approach; requires guidance(1.5)	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms(2)	Improves time complexity with minor gaps in efficiency(1.5)	Limited improvement in time complexity(1)	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions(2)	Shows reasonable analysis with some justification(1.5)	Limited analysis; weak reasoning(1)	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation(1.5)	Code is mostly clear with minor documentation gaps(1)	Code lacks structure and proper documentation(0.5)	Poorly written code with no documentation

	Problem Identification (20%)	Debugging (25%)	Optimization (20%)	Analysis (20%)	Code Quality (15%)	
Q1	1.5	2	1.5	1	1	7
Q2	1	2.5	1.5	1	1	7
Q3	1.5	2	1	1.5	1	7
Q4	1	1.5	2	2	1.5	8
Q5	2	1	2	1.5	1.5	8
						37

1. **A network administrator has configured a workstation with the following settings:**

- **IP Address: 192.168.10.65**
- **Subnet Mask: 255.255.255.192 (/26)**
- **Default Gateway: 192.168.10.1**

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- **Identifying the subnet to which the configured IP address belongs.**
- **Determining the network address, broadcast address, and valid host range.**
- **Verifying whether the configured IP address and gateway are valid.**
- **Identifying the configuration errors.**
- **Optimizing the configuration by assigning an appropriate IP address and default gateway.**
- **Calculating the total number of usable host addresses after correction.**

Given

- IP Address = **192.168.10.65**
- Subnet Mask = **255.255.255.192 (/26)**
- Configured Gateway = **192.168.10.1**
- Required subnet = **192.168.10.0/26**

Step 1: Find the subnets

For /26: **Subnet mask = 255.255.255.192**

The interesting octet is the last octet.

Calculate the block size: **256 − 192 = 64**

So the subnets increase by 64:

Subnet	Network Address	Broadcast Address
1	192.168.10.0	192.168.10.63
2	192.168.10.64	192.168.10.127
3	192.168.10.128	192.168.10.191
4	192.168.10.192	192.168.10.255

Step 2: Find where 192.168.10.65 belongs

The IP is: **192.168.10.65**

Look at the last octet: **65**.

65 falls between **64 and 127**.

Therefore: **192.168.10.65 belongs to the subnet 192.168.10.64/26.**

Its details are:

- **Network address:** 192.168.10.64
- **Broadcast address:** 192.168.10.127
- **Valid host range:** 192.168.10.65 – 192.168.10.126

Step 3: Check whether the IP address is valid

Configured IP: **192.168.10.65**

Valid host range is: **192.168.10.65 – 192.168.10.126**

So: **192.168.10.65 is a valid host IP.**

It is actually the **first usable host** in the 192.168.10.64/26 subnet.

Step 4: Check the gateway

Configured gateway:

192.168.10.1

But the workstation is in:

192.168.10.64/26

Valid hosts in this subnet are:

192.168.10.65 – 192.168.10.126

The gateway **192.168.10.1** belongs to:

192.168.10.0/26

because:

- Network = 192.168.10.0
- Hosts = 192.168.10.1 – 192.168.10.62
- Broadcast = 192.168.10.63

Therefore: **192.168.10.1 is NOT a valid gateway for the 192.168.10.64/26 subnet.**

Step 5: Identify the errors

There are actually **two different subnets involved**.

Intended subnet

The question says the workstation should communicate with systems in: **192.168.10.0/26**

That subnet has:

- Network: **192.168.10.0**
- Hosts: **192.168.10.1 – 192.168.10.62**
- Broadcast: **192.168.10.63**

But the workstation has: **192.168.10.65**

which belongs to: **192.168.10.64/26**

So the main error is: **The IP address is configured for the wrong subnet.**

The gateway 192.168.10.1 is actually correct **for the intended 192.168.10.0/26 subnet**, but it is incorrect for the currently configured IP 192.168.10.65.

Step 6: Correct the configuration

Since the workstation needs to communicate with systems in:

192.168.10.0/26

we should give it an IP from the valid host range:

192.168.10.1 – 192.168.10.62

But don't use 192.168.10.1 because that is the gateway.

For example, assign:

IP Address = 192.168.10.2

Subnet Mask = 255.255.255.192

Default Gateway = 192.168.10.1

So the corrected configuration is:

Setting	Correct Value
IP Address	**192.168.10.2**

Setting	Correct Value
Subnet Mask	**255.255.255.192 (/26)**
Default Gateway	**192.168.10.1**
Network Address	**192.168.10.0**
Broadcast Address	**192.168.10.63**

Now both the workstation and gateway are in the **same subnet**.

Step 7: Calculate usable host addresses

For /26: IPv4 has 32 bits.

Host bits: **32 − 26 = 6 host bits**

Total addresses: **2^6 = 64**

Two addresses cannot be assigned to hosts:

- Network address
- Broadcast address

Therefore: **Usable hosts = 64 − 2 = 62**

Answer: 62 usable host addresses

Final answer

Item	Answer
Configured IP	192.168.10.65
Subnet Mask	255.255.255.192 (/26)
Subnet of configured IP	**192.168.10.64/26**
Network Address	**192.168.10.64**
Broadcast Address	**192.168.10.127**
Valid Host Range	**192.168.10.65 – 192.168.10.126**
Is 192.168.10.65 valid?	**Yes**
Configured Gateway	192.168.10.1
Is gateway valid for 192.168.10.65?	**No**
Intended subnet	**192.168.10.0/26**
Correct IP example	**192.168.10.2**
Correct Gateway	**192.168.10.1**
Usable hosts after correction	**62**

2. Two devices are configured with the following IPv6 addresses:

- **Device A: 2001:db8::ff00:42:8329/64**
- **Device B: 2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- **Expanding both IPv6 addresses into their complete notation.**
- **Determining the network prefix of each device.**
- **Verifying whether both devices belong to the same subnet.**
- **Identifying the addressing or prefix mismatch.**
- **Optimizing the IPv6 configuration by assigning appropriate addresses.**
- **Calculating the number of available host addresses within the corrected subnet.**

Given

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Step 1: Expand both IPv6 addresses

IPv6 has **8 groups**, and each group contains **4 hexadecimal digits**.

Device A

Given: 2001:db8::ff00:42:8329

:: represents the missing zero groups.

Expanded form: **2001:0db8:0000:0000:0000:ff00:0042:8329**

So: **Device A = 2001:0db8:0000:0000:0000:ff00:0042:8329/64**

Device B

Given: 2001:db8:0:0:1::1

Expanded form: **2001:0db8:0000:0000:0001:0000:0000:0001**

So: **Device B = 2001:0db8:0000:0000:0001:0000:0000:0001/64**

Step 2: Find the network prefix

The prefix is /64.

That means:

First 4 groups = Network Prefix

Last 4 groups = Interface ID

Device A

Full address: 2001:0db8:0000:0000:0000:ff00:0042:8329

Divide it: 2001:0db8:0000:0000 | 0000:ff00:0042:8329

Therefore: **Network prefix of A = 2001:0db8:0000:0000::/64**

or simply: **2001:db8::/64**

Device B

Full address: 2001:0db8:0000:0000:0001:0000:0000:0001

Divide it: 2001:0db8:0000:0000 | 0001:0000:0000:0001

Therefore: **Network prefix of B = 2001:0db8:0000:0000::/64** or: **2001:db8::/64**

Step 3: Are they in the same subnet?

Compare the first 4 groups:

Device A

2001:0db8:0000:0000

Device B

2001:0db8:0000:0000

They are **exactly the same**.

Therefore: **Both devices belong to the same /64 subnet.**

The common subnet is: **2001:db8::/64**

Important: There is NO prefix mismatch

This question says communication fails and asks you to identify the prefix mismatch.

But mathematically, **there is no prefix mismatch in the addresses given**.

Device A:

2001:db8::ff00:42:8329/64

belongs to: **2001:db8::/64**

Device B:

2001:db8:0:0:1::1/64

also belongs to: **2001:db8::/64**

The 1 in Device B is in the **5th group**, which is part of the host/interface portion when the prefix is /64.

Therefore: It is **incorrect** to say that Device B is in 2001:db8:0:0:1::/64.

Because /64 only considers the **first four groups** as the network prefix.

Step 4: Optimize the configuration

Since both addresses already belong to the same /64 subnet, **no IP address correction is required based on the information given**.

The existing configuration is valid:

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Both belong to: **2001:db8::/64**

If the intention is to use simpler addresses, you could configure:

Device A: 2001:db8::1/64

Device B: 2001:db8::2/64

Both are clearly within: **2001:db8::/64**

and have unique interface IDs.

Step 5: Calculate available host addresses

IPv6 has **128 bits**.

Prefix length = **64 bits**

Therefore, host/interface bits: **128 − 64 = 64 bits**

Number of possible addresses: $\mathbf{2^{64}}$

Therefore: $\mathbf{2^{64}}$ **= 18,446,744,073,709,551,616**

So the /64 subnet contains: **18,446,744,073,709,551,616 possible IPv6 addresses**

Unlike IPv4, IPv6 does **not** normally subtract network and broadcast addresses. IPv6 has no broadcast address.

Final answer

Task	Answer
Device A expanded	**2001:0db8:0000:0000:0000:ff00:0042:8329/64**
Device B expanded	**2001:0db8:0000:0000:0001:0000:0000:0001/64**
A network prefix	**2001:db8::/64**
B network prefix	**2001:db8::/64**
Same subnet?	**Yes**
Prefix mismatch?	**No**
Correct subnet	**2001:db8::/64**
Possible optimized A	**2001:db8::1/64**
Possible optimized B	**2001:db8::2/64**
Host/interface bits	**64 bits**
Available IPv6 addresses	$\mathbf{2^{64}}$ **= 18,446,744,073,709,551,616**

3. An organization has allocated the network 192.168.1.0/24 into several departments using the
subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others
have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- **Identifying the network address, broadcast address, and host range for each subnet.**
- **Calculating the number of usable and unused IP addresses.**
- **Identifying inefficient subnet allocations.**
- **Debugging the addressing plan by locating subnet allocation mistakes.**
- **Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.**

1. Understand the three subnet sizes

For IPv4: **/26**

Subnet mask = **255.255.255.192**

Host bits = 32 − 26 = 6

Total addresses: **2^6 = 64**

Usable addresses: **64 − 2 = 62**

/27

Subnet mask = **255.255.255.224**

Host bits = 32 − 27 = 5

Total: **2^5 = 32**

Usable: **32 − 2 = 30**

/28

Subnet mask = **255.255.255.240**

Host bits = 32 − 28 = 4

Total: **2^4 = 16**

Usable: **16 − 2 = 14**

2. Analyze /26 subnet

Network: **192.168.1.0/26**

Block size: **256 − 192 = 64**

So:

- Network address = **192.168.1.0**
- First host = **192.168.1.1**
- Last host = **192.168.1.62**
- Broadcast = **192.168.1.63**
- Usable hosts = **62**

Other possible /26 subnets

Subnet	Network	Usable Host Range	Broadcast	Usable Hosts
1	192.168.1.0/26	.1 – .62	.63	62
2	192.168.1.64/26	.65 – .126	.127	62
3	192.168.1.128/26	.129 – .190	.191	62
4	192.168.1.192/26	.193 – .254	.255	62

3. Analyze /27 subnet

Block size: **256 − 224 = 32**

Possible /27 networks are: 0, 32, 64, 96, 128, 160, 192, 224

For example: **192.168.1.64/27**

- Network = **192.168.1.64**
- First host = **192.168.1.65**
- Last host = **192.168.1.94**
- Broadcast = **192.168.1.95**
- Usable hosts = **30**

Subnet	Network	Usable Host Range	Broadcast	Usable Hosts
1	192.168.1.0/27	.1 – .30	.31	30
2	192.168.1.32/27	.33 – .62	.63	30
3	192.168.1.64/27	.65 – .94	.95	30
4	192.168.1.96/27	.97 – .126	.127	30
5	192.168.1.128/27	.129 – .158	.159	30
6	192.168.1.160/27	.161 – .190	.191	30
7	192.168.1.192/27	.193 – .222	.223	30
8	192.168.1.224/27	.225 – .254	.255	30

4. Analyze /28 subnet

Block size: **256 − 240 = 16**

Possible networks: 0, 16, 32, 48, 64, 80, ...

For example: **192.168.1.128/28**

- Network = **192.168.1.128**
- First host = **192.168.1.129**
- Last host = **192.168.1.142**
- Broadcast = **192.168.1.143**
- Usable hosts = **14**

Subnet	Network	Usable Host Range	Broadcast	Usable Hosts
1	192.168.1.0/28	.1 – .14	.15	14
2	192.168.1.16/28	.17 – .30	.31	14
3	192.168.1.32/28	.33 – .46	.47	14
4	192.168.1.48/28	.49 – .62	.63	14
5	192.168.1.64/28	.65 – .78	.79	14
6	192.168.1.80/28	.81 – .94	.95	14
7	192.168.1.96/28	.97 – .110	.111	14
8	192.168.1.112/28	.113 – .126	.127	14
9	192.168.1.128/28	.129 – .142	.143	14
10	192.168.1.144/28	.145 – .158	.159	14
11	192.168.1.160/28	.161 – .174	.175	14
12	192.168.1.176/28	.177 – .190	.191	14

Subnet	Network	Usable Host Range	Broadcast	Usable Hosts
13	192.168.1.192/28	.193 – .206	.207	14
14	192.168.1.208/28	.209 – .222	.223	14
15	192.168.1.224/28	.225 – .238	.239	14
16	192.168.1.240/28	.241 – .254	.255	14

5. What is the inefficiency?

This is the main idea of the question.

Suppose a department has **10 computers**.

If you give it /26: **62 usable addresses**

Required = 10

Unused: **62 − 10 = 52**

That's a lot of wasted addresses.

A /28 gives:

14 usable addresses

Unused: **14 − 10 = 4**

Much better.

Example

Suppose departments require:

Department	Required Hosts
Administration	50
Finance	25
HR	12
Sales	6

Using the same subnet size for everyone would waste addresses.

Instead, VLSM gives each department a subnet based on its requirement.

6. Optimized VLSM solution

Start with the **largest department first**.

Administration → 50 hosts

Needs at least 50 usable addresses.

/26 gives 62 usable.

So:

Administration = 192.168.1.0/26

- Network = .0
- Hosts = .1 – .62
- Broadcast = .63
- Usable = **62**
- Required = 50
- Unused = **12**

Finance → 25 hosts

Needs 25.

/27 gives 30 usable.

Next available address is .64.

Finance = 192.168.1.64/27

- Network = .64
- Hosts = .65 – .94
- Broadcast = .95
- Usable = **30**
- Required = 25
- Unused = **5**

HR → 12 hosts

Needs 12.

/28 gives 14 usable.

Next available address is .96.

HR = 192.168.1.96/28

- Network = .96
- Hosts = .97 – .110
- Broadcast = .111
- Usable = **14**
- Required = 12
- Unused = **2**

Sales → 6 hosts

Needs 6.

A /29 provides:

$2^{3} - 2 = 6$ usable addresses

So:

Sales = 192.168.1.112/29

- Network = .112
- Hosts = .113 – .118
- Broadcast = .119
- Usable = **6**
- Required = 6
- Unused = **0**

7. Final optimized allocation

Department	Required	CIDR	Network	Host Range	Broadcast	Usable	Unused
Administration	50	/26	192.168.1.0	.1 – .62	.63	62	12
Finance	25	/27	192.168.1.64	.65 – .94	.95	30	5
HR	12	/28	192.168.1.96	.97 – .110	.111	14	2
Sales	6	/29	192.168.1.112	.113 – .118	.119	6	0

This is **VLSM** because different departments receive different subnet masks.

8. How to identify subnet allocation mistakes

When debugging, check these three things:

Network addresses must fall on correct boundaries

For /26: 0, 64, 128, 192

For /27: 0, 32, 64, 96, 128, 160, 192, 224

For /28: 0, 16, 32, 48, 64, 80, ...

If someone assigns something like:**192.168.1.50/26**

that's not a network address.

The actual network is:**192.168.1.0/26**

Subnets must not overlap

For example: 192.168.1.0/26

uses .0 – .63.

So you cannot assign another subnet that also uses addresses inside .0 – .63.

Don't use network/broadcast addresses as hosts

For: **192.168.1.64/27**

- .64 = network
- .65 – .94 = hosts
- .95 = broadcast

4. Two devices are configured with the following IPv6 addresses:

- **Device A: 2001:db8::ff00:42:8329/64**
- **Device B: 2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- **Expanding both IPv6 addresses into their complete notation.**
- **Determining the network prefix of each device.**
- **Verifying whether both devices belong to the same subnet.**
- **Identifying the addressing or prefix mismatch.**
- **Optimizing the IPv6 configuration by assigning appropriate addresses.**
- **Calculating the number of available host addresses within the corrected subnet.**

Given:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

1. Expanded notation

- **Device A:** 2001:0db8:0000:0000:0000:ff00:0042:8329/64
- **Device B:** 2001:0db8:0000:0000:0001:0000:0000:0001/64

2. Network Prefix

For /64, the **first 4 groups** are the network prefix.

- Device A → 2001:db8:0:0::/64
- Device B → 2001:db8:0:0::/64

3. Same subnet?

Yes. Both devices belong to the same subnet: **2001:db8::/64**

4. Addressing/prefix mismatch

There is **no actual prefix mismatch**. The 1 in Device B is in the host/interface portion, not the network prefix.

So, if communication fails, the problem must be due to another configuration issue such as routing, interface, firewall, or link connectivity.

5. Optimized configuration

For simplicity, addresses can be assigned as:

- Device A → 2001:db8::1/64

- Device B → 2001:db8::2/64

Both belong to **2001:db8::/64**.

6. Available host addresses

IPv6 has 128 bits and /64 uses 64 bits for the network.

Host bits: **128 − 64 = 64**

Therefore: **2^{64} = 18,446,744,073,709,551,616 addresses**

Final

Corrected subnet: 2001:db8::/64

Usable IPv6 addresses: 2^{64} = 18,446,744,073,709,551,616

5. A router contains the following routing table.

Destination Network Next Hop

192.168.1.0/24 10.0.0.2

192.168.2.0/24 10.0.0.3

Default Route 10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- **Identifying the matching routing entry using the longest-prefix match.**
- **Determining the next-hop router selected for packet forwarding.**
- **Verifying whether the destination IP belongs to the intended subnet.**
- **Identifying any routing configuration errors.**
- **Optimizing the routing table to improve packet forwarding efficiency.**
- **Identifying the route that will be used if no matching destination exists.**

Given Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

1. Destination IP

Destination = **192.168.2.100**

Check the routing table:

- 192.168.1.0/24 → Not a match
- 192.168.2.0/24 → Match

Therefore, using **Longest Prefix Match**, the router selects: **192.168.2.0/24 → 10.0.0.3**

2. Next-hop router

The packet will be forwarded to: **10.0.0.3**

3. Verify destination subnet

192.168.2.100 belongs to: **192.168.2.0/24**

Valid host range: **192.168.2.1 – 192.168.2.254**

So the destination IP is valid.

4. Routing error

There is **no error in the routing table based on the information given**.

The correct route exists: **192.168.2.0/24 → 10.0.0.3**

If packets still fail, the problem may be with the next-hop router, interface, connectivity, or return route.

5. Optimization

The routing table is already efficient because it uses specific /24 routes and a default route.

A more specific route would take priority if needed because of **Longest Prefix Match**.

6. If no matching route exists

The router uses the: **Default Route → 10.0.0.1**